Updating parameters

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Updating parameters before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Gradients with a numerical check

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Loss functions

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Composing layers

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Gradient descent uses new parameter = old parameter - **learning rate** times its gradient entry. The minus sign chooses the locally downhill direction for a positive **learning rate**.

A negative gradient therefore increases the corresponding parameter.

Keep the Week 10 example: x = 2, target 3, y = 2w + b and L = (y - 3) squared. At [w, b] = [1, 0], the loss is 1 and gradient is [-4, -2].

With **learning rate** 0.1, new w = 1 - 0.1 times -4 = 1.4 and new b = 0 - 0.1 times -2 = 0.2.

Quantity	Before update	After update
Parameters [w, b]	[1, 0]	[1.4, 0.2]

Quantity	Before update	After update
Prediction	2	3
Squared loss	1	0

This particular step reaches zero loss on one example. It does not establish a unique parameter solution, successful learning on other examples or a generally ideal **learning rate**.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Updating parameters trace and record the first step you can explain without help.

Use W2: Detect **overshoot**ing for Updating parameters. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Updating parameters explanation against the supplied facts before discussing it.

Use the worked case for Updating parameters. Proposed explanation: “Subtracting a negative gradient decreases the parameter”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Update a new model independently

Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Apply a negative gradient

> Apply a negative gradient. Apply the supplied Updating parameters facts and keep your reasoning.

Starting with parameter a = 2, gradient -3 and **learning rate** 0.2, calculate the new a.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Detect overshooting

> Detect overshooting. Apply the supplied Updating parameters facts and keep your reasoning.

For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Preserve one snapshot

> Preserve one snapshot. Apply the supplied Updating parameters facts and keep your reasoning.

Explain why recomputing the bias gradient after changing w is not the **simultaneous update** specified here.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Update a new model independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Update a new model independently. Apply the supplied Updating parameters facts and keep your reasoning.

Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Evaluate a zero gradient

Evaluate a zero gradient. Apply the supplied Updating parameters facts and keep your reasoning.

At a point with gradient [0, 0], what does this update do? Does that alone prove a global minimum on any loss surface?

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Updating parameters to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled updating parameters **evidence** sheet to Image-pattern Network Investigation. Show how you can apply a **learning rate** to an ordered gradient.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

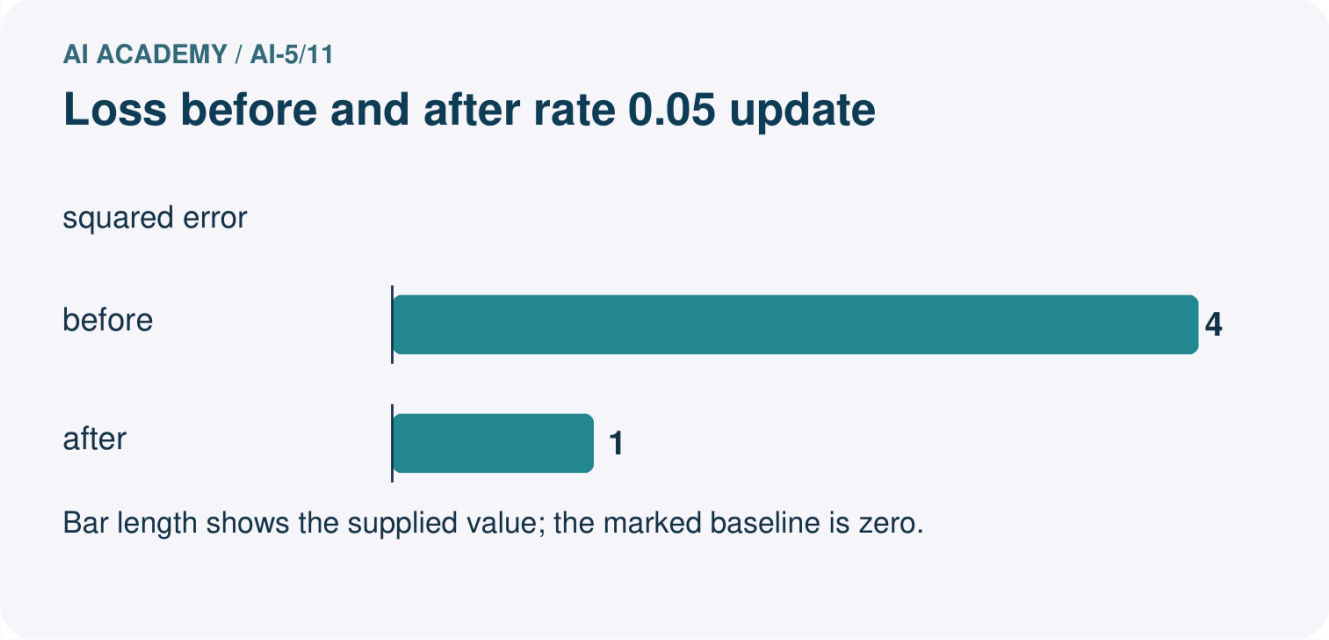
Pictorial case practice: Updating parameters

> Use one explicit gradient update and then recompute the loss. Explain the labelled step before checking the changed condition.

Use one explicit gradient update and then recompute the loss.

Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/11

Updating parameters

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gradient

Gradient for [w, b]=[2 x (-2) x 2, 2 x (-2)]=[-8,-4].



2. Update simultaneously

New w=1-0.05 x (-8)=1.4; new b=1.2.



3. Check new loss

New y=4, residual -1, loss 1: lower than 4.

CHANGE ONE THING

Set learning rate 0.2 from the same starting point.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use one explicit gradient update and then recompute the loss.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? New y=4, residual -1, loss 1: lower than 4.

1. Check new loss

2. Compute gradient

3. Update simultaneously

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check new loss New y=4, residual -1, loss 1: lower than 4. **Gradient descent** subtracts **learning rate** times gradient.

A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one. The small step brings the **prediction** closer to five, reducing squared error.

A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Set **learning rate** 0.2 from the same starting point. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: New w=2.6, b=1.8, y=7, squared loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

Repair a missing building block

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

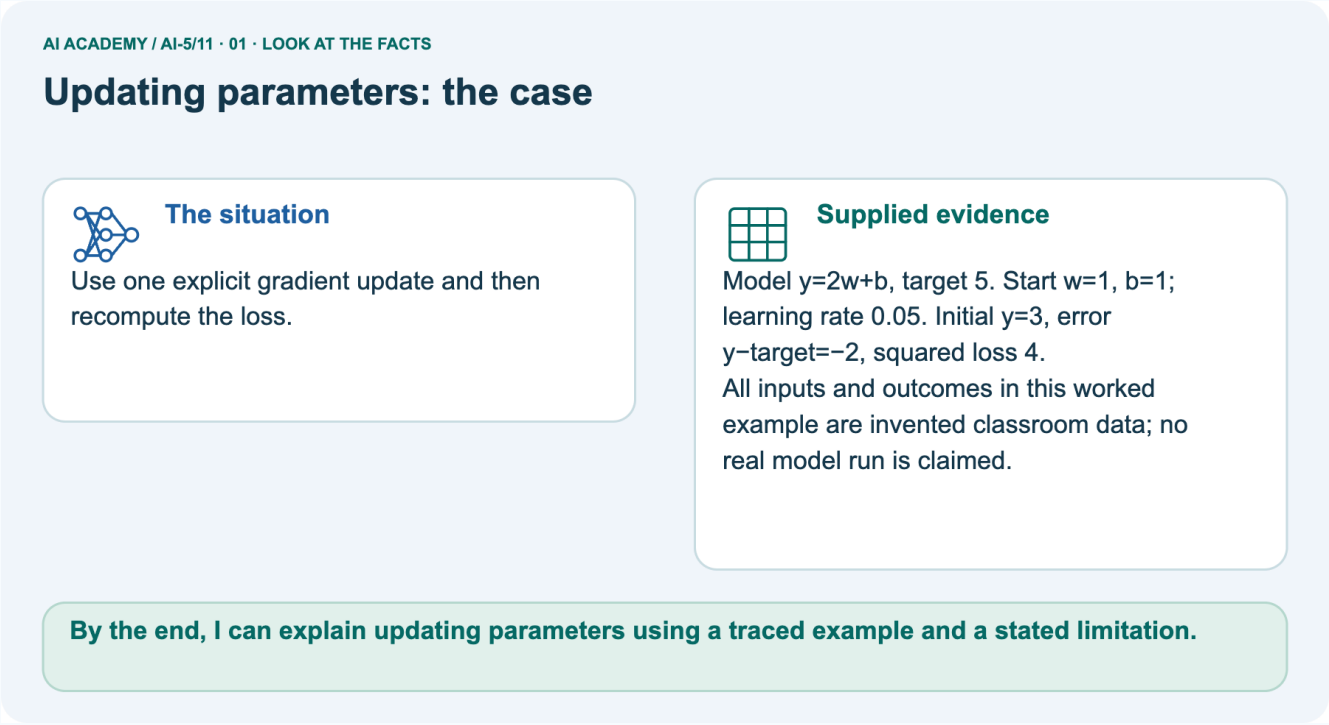
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Updating parameters picture from its labels, then record a relevant limit.

01. Updating parameters: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

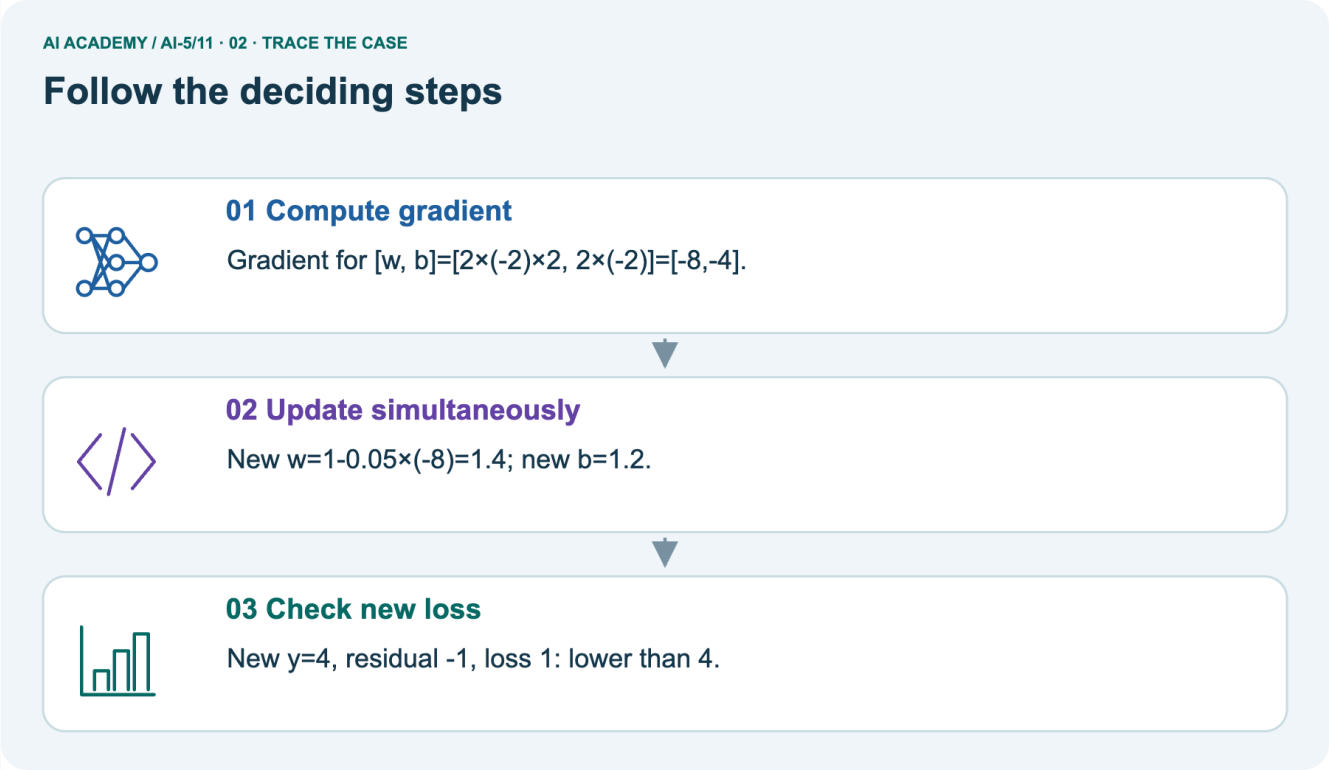
Use one explicit gradient update and then recompute the loss. Supplied facts: Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



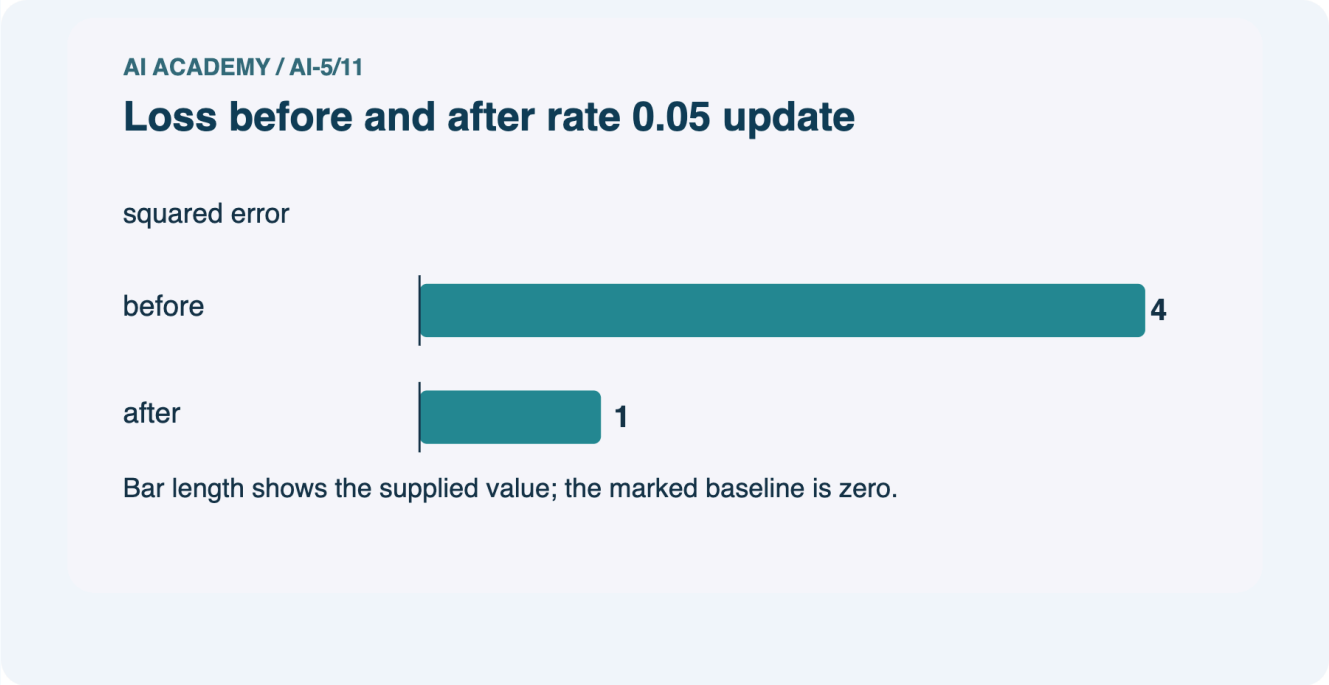
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute gradient: Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4]. → Update simultaneously: New w=1-0.05×(-8)=1.4; new b=1.2. → Check new loss: New y=4, residual -1, loss 1: lower than 4.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Loss before and after rate 0.05 update



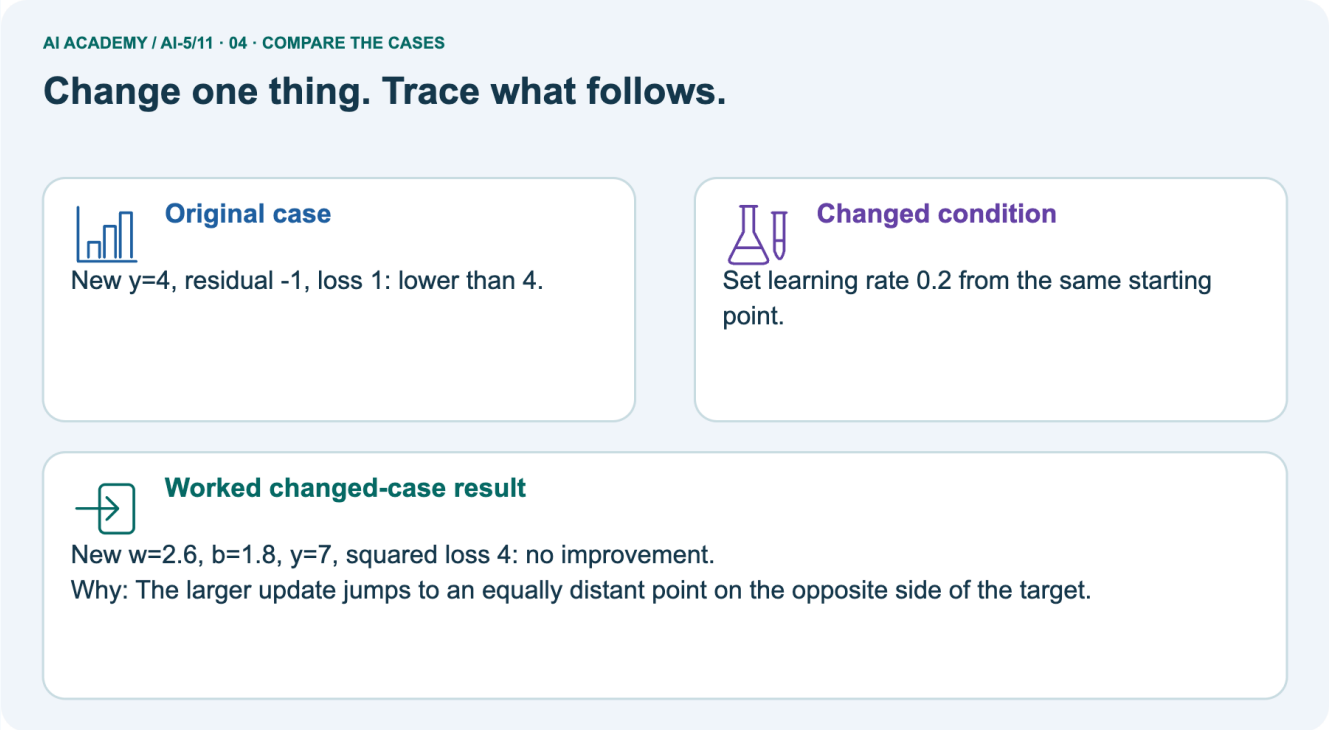
Follow the labelled picture with the matching explanation. Open the vector diagram

Loss before and after rate 0.05 update. Units: squared error. before: 4. after: 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

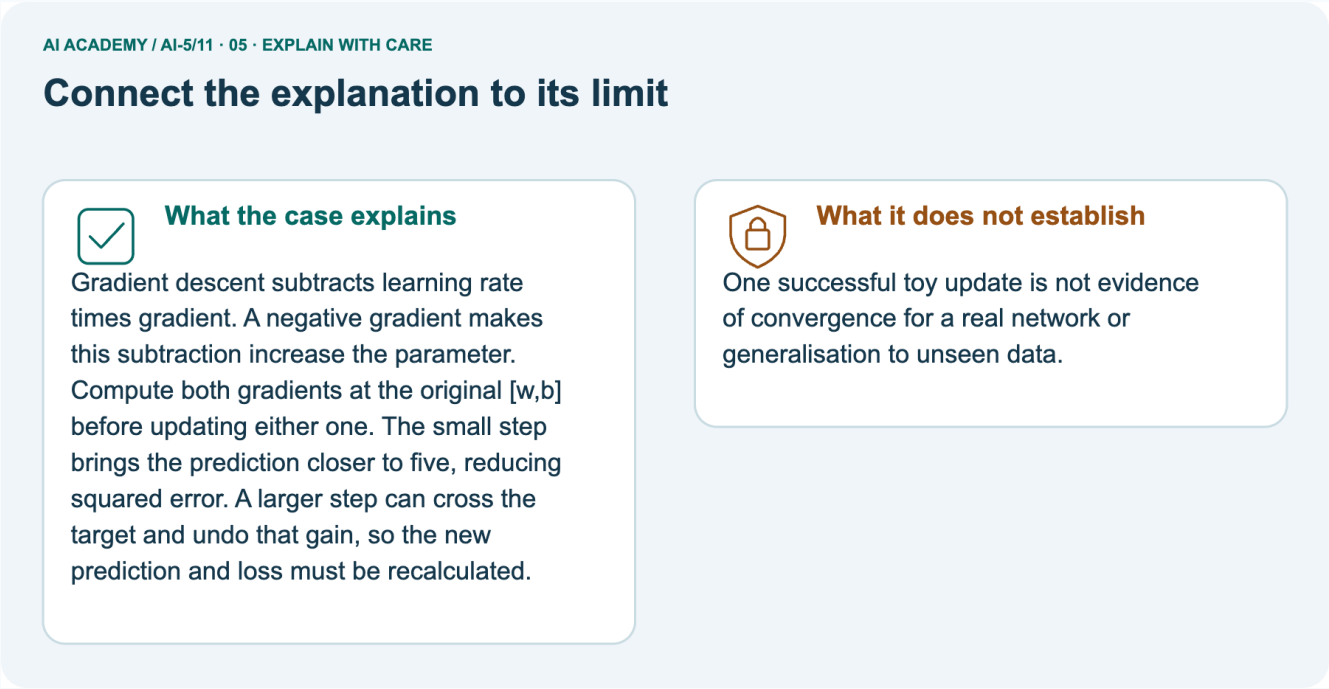
Original case: New y=4, residual -1, loss 1: lower than 4. Changed condition: Set **learning rate** 0.2 from the same starting point. Expected result: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Gradient descent subtracts **learning rate** times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one.

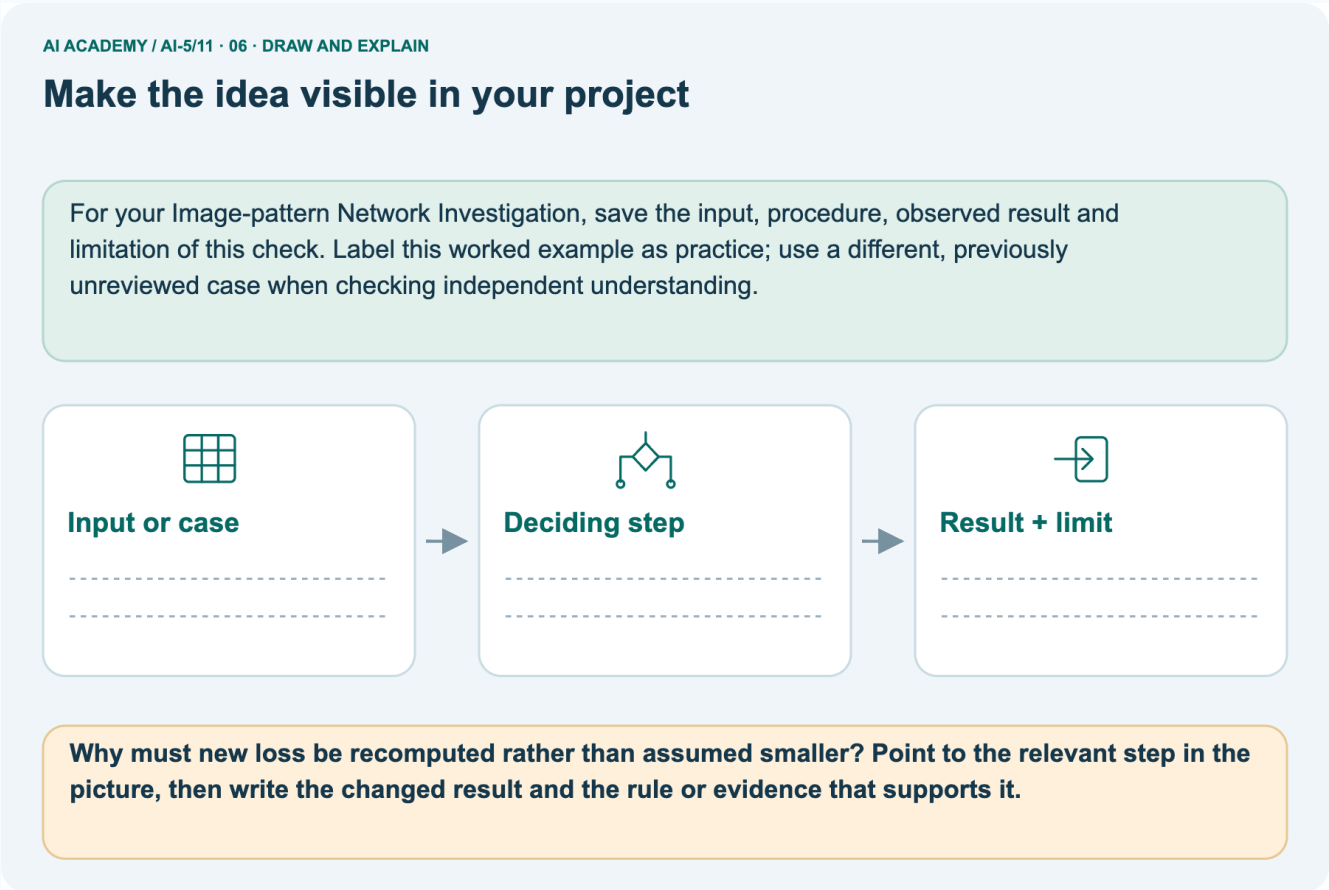
The small step brings the **prediction** closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

Limit: One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.